

EVC Parallel Function



Function Purpose

1. To realize orderly charging of two EV Charger by centralized management of the remaining PV and remaining household electricity capacity assigned to the EV Charger.
2. Avoid disordered charging leading to equipment logic confusion and electricity consumption fluctuations.
3. Through the management of EV Charger parallel solution, multiple EV Chargers can be effectively integrated with resources, and the efficiency of EV Chargers can be improved to meet the needs of many different electric vehicle owners.

Function Description

Primary EV charger obtains the surplus PV information through communication with the inverter (or CT sampling). It also controls the working current of the two EV charger according to the surplus PV information, the working status (charging/non-charging) and mode (Green/ECO/Fast) of the two EV charger and the household current limit value (main breaker).

Scenario

1. Photovoltaic Scenario (with Inverter)

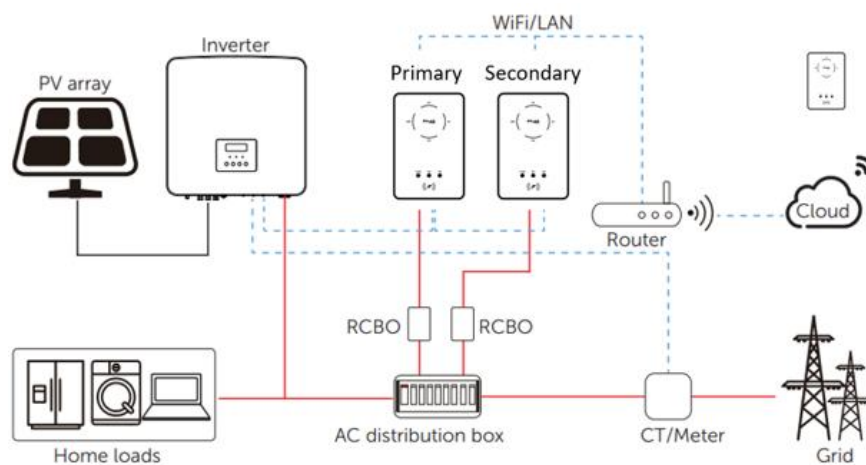


Figure 1 With Inverter

Tips:

1. Compatible Version: Home edition and Fusion edition.
2. Both EV charger need to be of the same model and same rated power.
3. Inverters need to be compatible.

2. Photovoltaic Scenario (with CT)

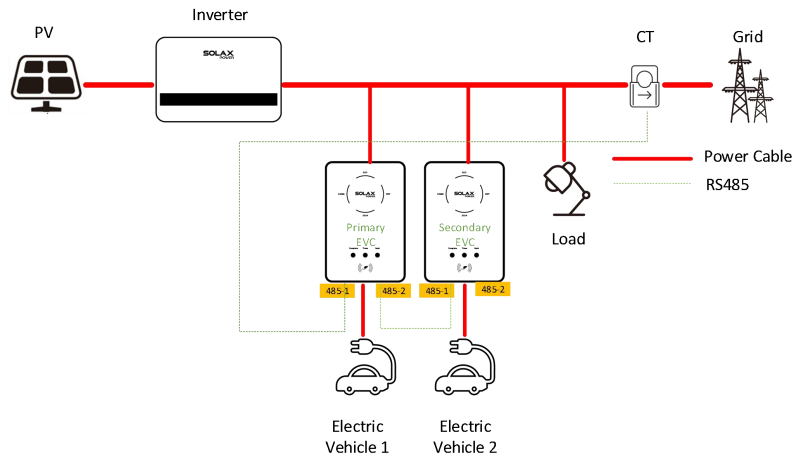


Figure 2 With CT

Principle: Primary EVC analyses the current excess PV information to control the operating current of the EVC by obtaining the current information of the feed-in current at the grid-connected point through the CT.

Adaptability: In this scenario, the inverter model can be any model that is not compatible with the EV charger and also inverters from other companies.

Note: In scenarios where the PV system has power limitations, the Green mode of the EV charger can only be used properly when connected to an inverter.

Reason: The inverter can give information about whether there is PV left or not to the EV charger but the CT can't.

3. Photovoltaic Scenario (With Meter)

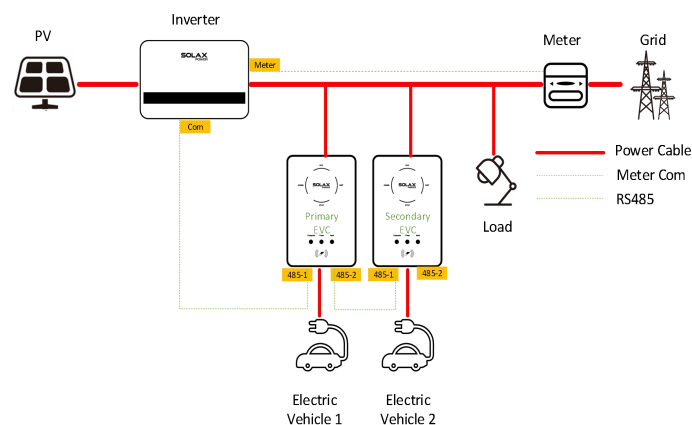


Figure 3 With Meter

Adaptability: In this scenario, the EV charger can only be Home version and Fusion version.

4. Small Commercial and Industrial Scenario (Datahub centralizes dynamic load balancing for small charging stations, adjusting charger power based on real-time status and capacity data.)

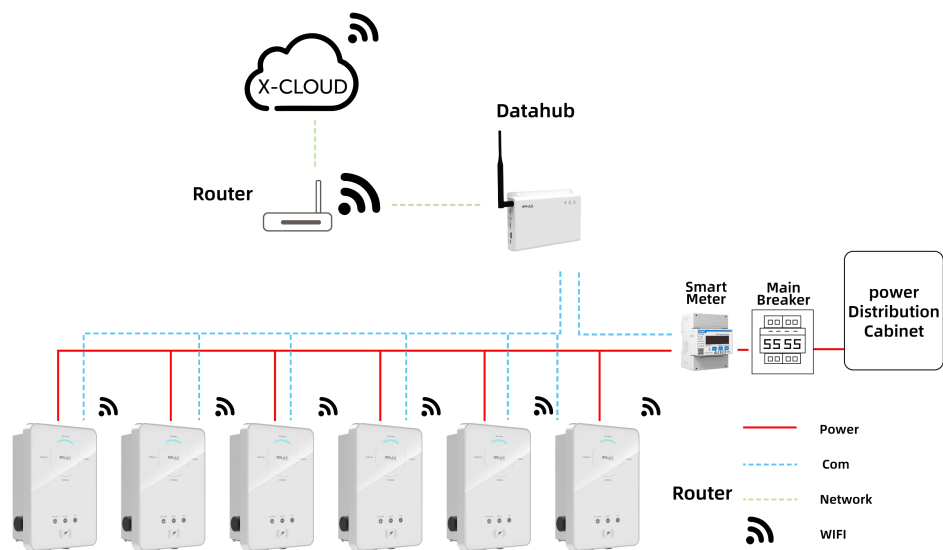


Figure 5 Parallel with Datahub

Tips:

1. Compatible Version: Commercial version and Fusion version.
2. Both EV chargers need to be of the same model and same rated power.

Wiring: Set Up

1. Photovoltaic Scenario

With Inverter: Primary EV Charger's 485-1 communicates with inverter's com port. **Primary EV charger's 485-2 port** connects with **secondary EV Charger's 485-1 port**. The inverter com port connects with EV charger, address is 70, baud rate is 9600.

With CT: Primary EV Charger's RJ45 port (485-1) is connected to the CT. Primary EV Charger's 485-2 is connected to Secondary's 485-1.

Port definition and APP settings

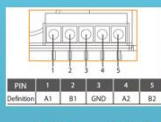
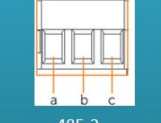
Model	Interface Board	RJ45 Terminal (485-1: pin4&5)	Green Terminal	App Setting
Home Version				• Path: Setting-Advanced settings-Parallel setting • Setting: enable/disable and parallel setting(1:1,1:2,2:1) • The setup is done only on the Primary master, and the slave is automatically recognised for control after the master is set up. The allocation ratio is for the remaining PV and the remaining home power capacity.
Mix Version				

Figure 7 Port definition and APP settings

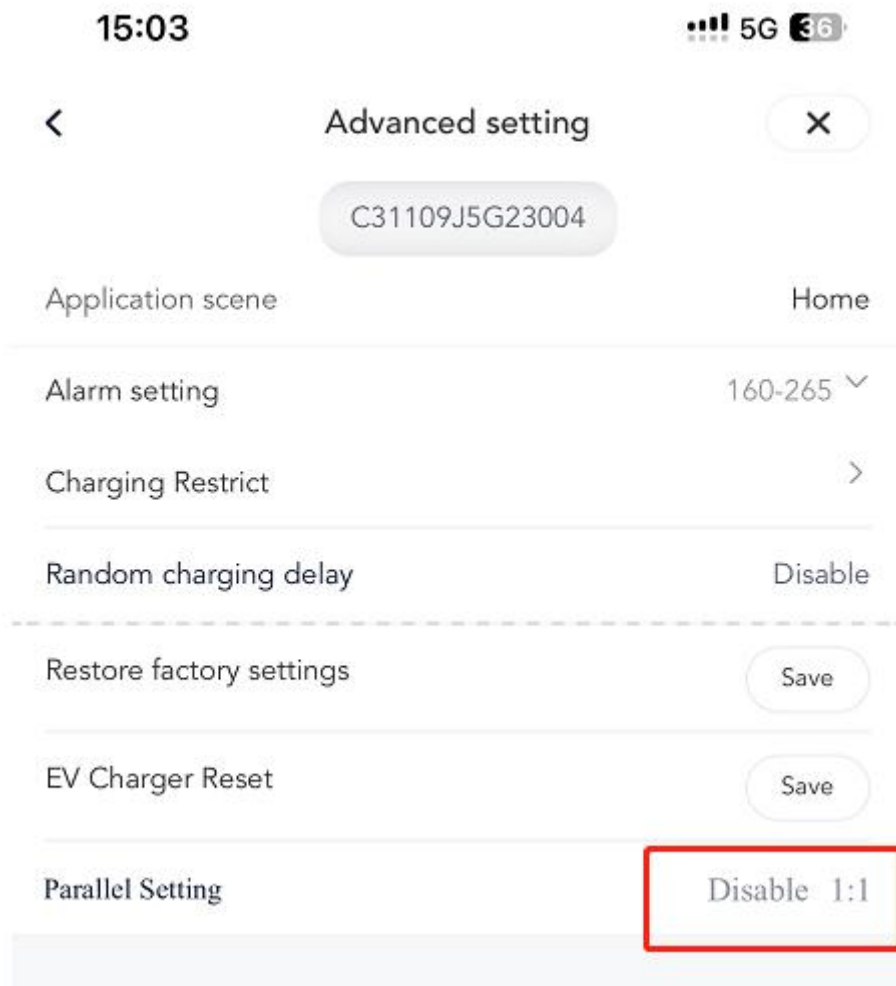


Figure 8 Parallel Setting (EV Charger)

When both EV Chargers satisfy the working conditions, the allocation is done proportionally. When only one EV charger can satisfy the working conditions, the EV Charger with high allocation ratio will be given priority to satisfy the EV Charger with high allocation ratio, so as to ensure that the remaining PV is not discarded but is fully utilized.

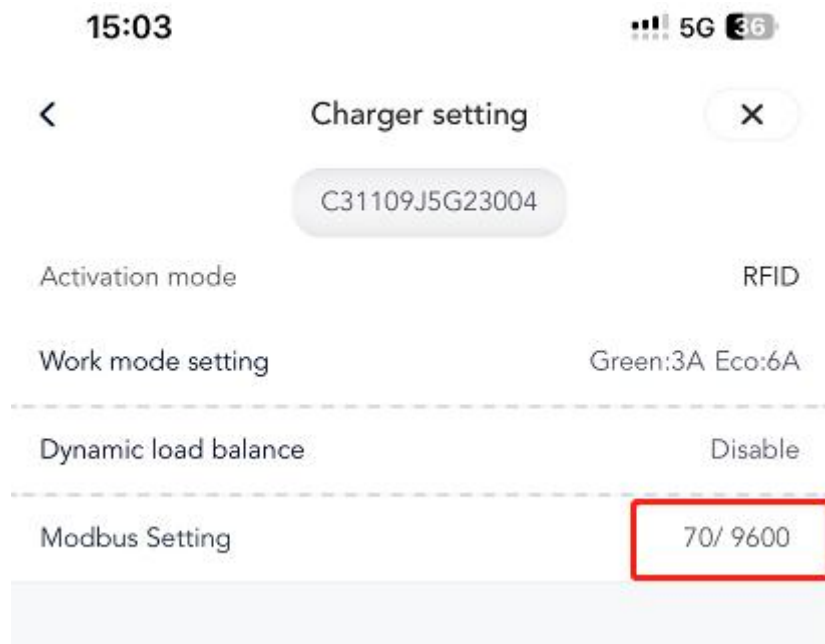


Figure 9 Parallel Address and Baud Rate Setting (EV Charger)

When we use the EV Charger in parallel, we need to set the baud rate and address of the EV Charger on the master side as well as on the slave side.

2. Small Commercial and Industrial Scenario

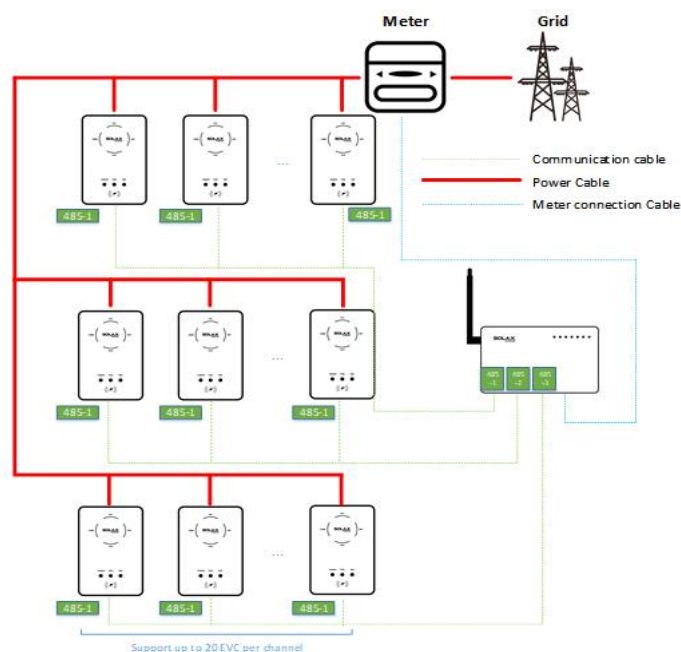


Figure 11 Wiring of Parallel with Datahub

Wiring: Datahub designates an RS485 port to connect to the meter and sets the corresponding interface to connect the device type to the meter and sets the number of devices and initial address. The other 485 port of Datahub is connected to the 485-1 of EVC, and the 485-1 of EVC is connected hand in

hand in series. And set the corresponding port to connect the device type as EVC and set the number of devices and initial address. EVC need to set the address, the default address is 70, the following EV charger in turn plus 1 to set.

Datahub Setting:

RS485 Channel	Device Type	Initial Address	Number of Devices
1	EV Charger	70	
2	EV Charger	75	10
3	EV Charger	85	5
4	Meter	1	1

Check Device Detail

Save

Figure 12 Datahub Setting

Datahub: No need to set the address of the devices themselves one by one, directly click on the automatic addition of devices.

Datahub will automatically assign the address to the connected devices.

The commercial version does not support the automatic address allocation.

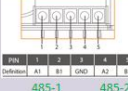
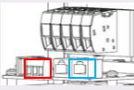
Models	Interface Board	RJ45 Terminal (485-1; Pin4&5)	Green Terminal	APP Setting																		
Commercial Edition		<table><tr><th>PIN</th><th>1</th><th>2</th><th>3</th><th>4</th><th>5</th><th>6</th><th>7</th><th>8</th></tr><tr><td>Definition</td><td>L1, CT+</td><td>L1, CT-</td><td>L2, CT+</td><td>A1</td><td>B1</td><td>L2, CT-</td><td>L3, CT+</td><td>L3, CT-</td></tr></table>	PIN	1	2	3	4	5	6	7	8	Definition	L1, CT+	L1, CT-	L2, CT+	A1	B1	L2, CT-	L3, CT+	L3, CT-	 485-1 485-2	Path: Setting----Slave address
PIN	1	2	3	4	5	6	7	8														
Definition	L1, CT+	L1, CT-	L2, CT+	A1	B1	L2, CT-	L3, CT+	L3, CT-														
Fusion Edition		<table><tr><th>PIN</th><th>1</th><th>2</th><th>3</th><th>4</th><th>5</th><th>6</th><th>7</th><th>8</th></tr><tr><td>Definition</td><td>L1, CT+</td><td>L1, CT-</td><td>L2, CT+</td><td>A1</td><td>B1</td><td>L2, CT-</td><td>L3, CT+</td><td>L3, CT-</td></tr></table>	PIN	1	2	3	4	5	6	7	8	Definition	L1, CT+	L1, CT-	L2, CT+	A1	B1	L2, CT-	L3, CT+	L3, CT-	 a b c 485-2	Path: Setting----Charger settings----Modbus setting
PIN	1	2	3	4	5	6	7	8														
Definition	L1, CT+	L1, CT-	L2, CT+	A1	B1	L2, CT-	L3, CT+	L3, CT-														

Figure 13 Port definition and app-side settings